CalendarBot: from a pipeline to four agents

Generative Artificial Intelligence (580694) · Universidad de Concepción · Spring 2026
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REPO github.com/maty20arriagada/SML-GEN-AI-for-Calendar-services | PROMPTS one versioned file per agent in `deliverable3/prompts/` | TRACES every input, `resultados/traces/` | VIDEO open link (3 min)

3% → 74% → 92% — all 160 phrases correct: direct prompt → D2 solution → agent system

65% → 81% — independent set C (26, written blind): 5 fixed, 1 broken vs. D2; $p = 0.219$, a trend

49% → 93% — set B (41), one group per failure class of D2: D2 → agents

−27 / −8 — phrases lost when A3's tools / A4's audit are removed: the two pieces that carry the gain

01 The task, the failure and the criterion we kept

RUBRIC · CONTINUITY WITH D1–D2

SAME TASK, SAME CRITERION

A Spanish/English chat phrase becomes a 9-field calendar event. *Correct* = every scored field matches the ground truth (`evaluar.py`, 8 fields; `recurrence` not scored), `null`/`[]` when absent, nothing invented; relative dates from 2026-09-01. Unchanged since D1.

THE FAILURE D2 LEFT OPEN

D2 (Qwen3-4B + LoRA + date resolver) reached 88% on its own benchmark but still broke on three families:

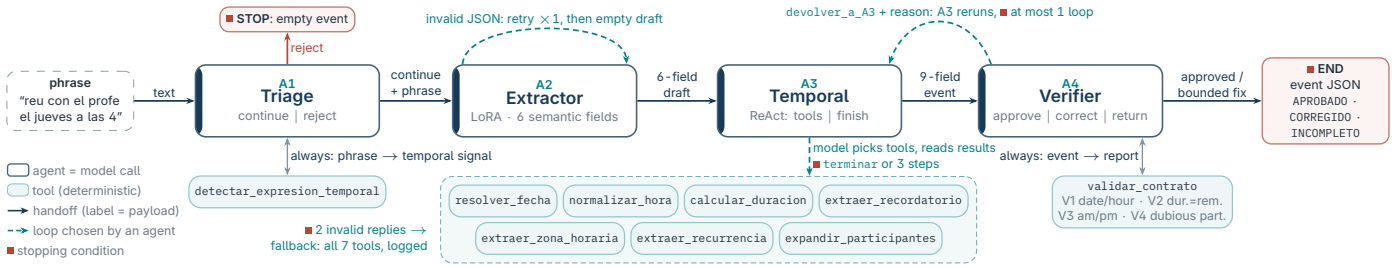
- reminder read as duration — "avísame 15 min antes"
- dates, hours, time zones — "31/09", "viernes 18", "7 pm", "horario de Madrid"
- participants invented — "ensayo de la banda" → `["banda"]`

DECLARED DEVIATIONS

Four agents on the *same* open model (Qwen3-4B-Instruct + LoRA `checkpoint-300`, merged). 160 phrases instead of 93: **A** (93, the D2 benchmark, where errors were diagnosed: optimistic), **B** (41, validation by failure class built after the error analysis, not independent), **C** (26, written blind).

02 Four agents, one model: who decides what happens next

RUBRIC · AGENT GRAPH



03 Why four roles: the work is split by the kind of error, not by the code

RUBRIC · AGENT DESIGN

	Role the task needs, and why no other agent covers it	Receives → returns · prompt file	Stops / hands off
A1	*Is this an event at all?* The only agent that can end a run, so an unusable reply means continue.	phrase + temporal signal → `{accion, razon, idioma}` `a1_triage_v2.txt`	reject: empty event; else → A2
A2	*Semantic fields*: what the D2 LoRA already does well. It sees no dates, so its habits cannot corrupt them.	phrase → 6-field JSON `a2_extractor_v1.txt`	valid → A3; invalid: retry once
A3	*Time reasoning.* Tools resolve dates and pm hours where a 4B model slips; only the model knows which ones a phrase needs.	phrase, A2 draft, ref. date, A4 objection → `{pensamiento, accion, tools}` per step `a3_agente_v1.txt`	`terminar`, 3 steps or 2 invalid replies → A4
A4	*Independent audit.* A model misses its own errors; a second reading against phrase and contract catches them. Only A4 can send work back.	phrase, event, `validar_contrato` report → `{razonamiento, accion, correcciones}` `a4_agente_v1.txt`	approve / correct → END; return → A3, once

04 What each technique buys: removed one at a time, traced to the failure it targets

RUBRIC · PROMPTING TECHNIQUES

Technique · agent	Failure it targets	Effect of removing it: all (C)
T1 tool use, model picks tools · A3	dates, hours, time zones	**−27** (C −1)
T2 contrastive few-shot · A4	participants invented	**0** (C 0)
T3 rubric + step-by-step · A4	participants invented · reminder read as duration	**0** (C 0)
T4 verifier + return loop · A4	participants invented A3's blind spots	**−8** (C −2)
T5 triage gate · A1	non-events, malformed input	**0** (C 0)

Reading. Two techniques carry the gain. Without **T1**, 37 outputs change and the date classes slip (weekday + number 4/4 → 2/4). Without **T4**, title complement falls from 6/6 to 2/6: the verifier is what keeps "ensayo de la banda" from inventing a participant. **T2, T3, T5 measure zero**, and the traces say why: A1 never rejected (no phrase is a non-event) and 45 of A4's 55 corrections hit a field it may not fix, so A4's wording rarely reaches the output. Loop A4→A3 in 4% of runs; A3 fallback 0%.

SET B, CORRECT PER FAILURE CLASS

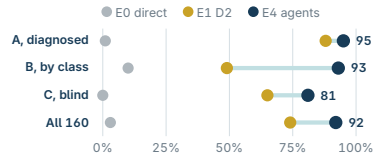
	E0	E1	E4	−T1	−T4
reminder read as duration	0/7	7/7	7/7	6/7	7/7
weekday + day number	0/4	0/4	4/4	2/4	4/4
impossible date (31/09)	4/4	1/4	4/4	4/4	4/4
English relative date	0/5	0/5	4/5	4/5	4/5
am/pm hour	0/5	2/5	5/5	4/5	5/5
title complement	0/6	2/6	6/6	6/6	2/6
compound role	0/5	3/5	3/5	2/5	3/5
control (no trap)	0/5	5/5	5/5	4/5	5/5

E0 direct prompt, E1 D2 solution, E4 agents, −Tk ablations; **gold** = the drops named in the reading. B was built after the error analysis: it shows *where* each piece acts, not how well the system generalises.

05 Evidence: same 160 inputs, same criterion (all scored fields correct) for every arm

RUBRIC · EVIDENCE OF IMPROVEMENT

Arm Δ = vs. E4	A n=93	B n=41	C n=26	All n=160	ΔC	ΔAll
E0 direct prompt, base model	1 (1%)	4 (10%)	0 (0%)	5 (3%)	−21	−142
E1 D2 solution	82 (88%)	20 (49%)	17 (65%)	119 (74%)	−4	−28
E4 full agent system	**88 (95%)**	**38 (93%)**	**21 (81%)**	**147 (92%)**	ref.	ref.
E4 − T1 tools (A3)	68 (73%)	32 (78%)	20 (77%)	120 (75%)	−1	−27
E4 − T2 few-shot (A4)	88 (95%)	38 (93%)	21 (81%)	147 (92%)	0	0
E4 − T3 rubric/CoT (A4)	88 (95%)	38 (93%)	21 (81%)	147 (92%)	0	0
E4 − T4 verifier + loop	86 (92%)	34 (83%)	19 (73%)	139 (87%)	−2	−8
E4 − T5 triage (A1)	88 (95%)	38 (93%)	21 (81%)	147 (92%)	0	0



Paired test vs. D2 (exact McNemar). Independent set C: E4 fixes 5 phrases E1 failed and breaks 1, $p = 0.219$: *a trend, not proof* with n = 26. All 160: 31 fixed, 3 broken, $p < 0.001$; A is optimistic and B is not independent, so C is the number we stand behind.

HARDWARE (MEASURED) Colab free · Tesla T4 15.6 GB · fp16 | Qwen3-4B-Instruct + LoRA (merged) for every agent | peak VRAM 8.74 GB agents vs. 8.12 GB direct prompt (+0.62 GB)

LIMITS A1/A2 outputs are cached and shared by all arms, so ablations differ only downstream | only C is blind | n is small, one checkpoint | A1 untested on real non-events

06 Where it still breaks: a real failure, read from its trace

RUBRIC · FAILURE CASE

in	"Dinner with Tom at 7 pm New York time on October 9" (set C, id 20)
A1	`detectar_expresion_temporal` → `fecha 2026-10-09` ✓; `continuar`
A2→A3	`{titulo: Dinner, hora: 19:00, participantes: [Tom]}` ✓
A3 step 1	`normalizar_hora` → 19:00; `extraer_zona_horaria` → America/Santiago
A3 steps 2–3	`extraer_zona_horaria` again, same answer; ■ step cap. `resolver_fecha` never called
A4	`validar_contrato` → []; "timezone must be New York" → `corregir`: rejected
out	INCOMPLETO: `fecha=null`, `zona_horaria=America/Santiago` (6/8 fields)

WHY IT FAILS

The phrase carried every value; three gaps lined up. `extraer_zona_horaria` knows only Madrid and Argentina and *silently* returns the Santiago default instead of `null`. A3 trusted the tool over the phrase and spent its steps re-asking it, so the date tool was never called. A4 saw the wrong zone but chose `corregir` on a field outside its bounded fixes instead of returning the work to A3: no invented value got in, but the loop that would have caught it was not taken. Trace: `traces/falla/`; video inputs: `traces/video/`.